

Tahsin Tajwar Tanni

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[Portfolio](#) | [LinkedIn](#) | [Github](#) | [Google Scholar](#)

PERSONAL PROFILE

- ❖ Computer Science and Engineering graduate with a thesis focused on pretrained language model security, specializing in defense mechanisms against fine-tuning attacks and improving model robustness under adversarial conditions.
- ❖ Skilled in deep learning research and experimentation using PyTorch and Hugging Face Transformers, full-stack web development with React, Node.js, and MongoDB, and reproducible research workflows in Python.

EDUCATION

01/2022 – 02/2026	BRAC UNIVERSITY BSc in Computer Science and Engineering	GPA: 3.60 / 4.00
03/2018 – 11/2020	<i>Govt. Haji Muhammad Mohsin College</i> Higher Secondary Certificate (HSC)	GPA: 5.00 / 5.00
01/2004 – 12/2018	<i>Bangladesh Mohila Samity Girls' High School and College</i> Secondary School Certificate (SSC)	GPA: 5.00 / 5.00

PROFESSIONAL EXPERIENCE

05/2025 – Present	AI Automation Trainee <i>180 RE, Toronto, Canada</i> • Built LLM-powered workflows using OpenAI and Anthropic APIs, automating multi-step business processes via n8n and third-party service integrations. • Developed and deployed AI-powered chatbots, integrating them into websites to automate user interactions and supports
02/2023 – 07/2024	Junior Web Developer <i>Weabers, Chattogram, Bangladesh</i> • Built responsive web applications; collaborated on API integration and feature deployment.

VOLUNTEERING

02/2025 - 07/2025	Python Instructor <i>Zentorra</i> • Taught foundational Python programming concepts to beginner-level learners.
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RESEARCH EXPERIENCE

01/2025 – 01/2026	Undergraduate Researcher <i>BRAC University, Department of Computer Science and Engineering</i> 01/2025-01/2026 • Investigated post-deployment vulnerabilities in pretrained language models, focusing on weight noise injection and low-precision quantization attacks that degrade model stability without altering architecture. • Designed and implemented Anchor-Guided Repair, a fine-tuning defense that jointly optimizes language modeling loss and anchor regularization to constrain parameter drift from a clean task-adapted baseline.
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- Evaluated the method across multiple LLM architectures and attack types (Gaussian noise, low-bit quantization), consistently recovering performance close to the clean baseline.

PUBLICATION

Rohan, A. M., Khan, N., **Tanni, T. T.**, Fardin, F., & Bushra, A. (2026). *Anchor-Guided Repair: A Defense Mechanism for Enhancing Stability of Compromised Pretrained Language Models Against Low-Precision and Weight Noise Attacks*. In *Proceedings of IndabaX Nigeria 2026: Building Scalable AI That Works: From Research to Deployment in Resource-Constrained Environments. Proceedings of Machine Learning Research*, **319**, 368–381. <https://proceedings.mlr.press/v319/>

ONGOING RESEARCH

Phantom Guard: A Fine-tuned Multi-Agent System for Robust Offline Slopsquatting Detection (*Role: Researcher*)

- Designing a dual-LLM generator-reviewer pipeline to detect hallucinated and slopsquatted package recommendations.
- Implementing fallback validation against official package registries and repository APIs.
- Exploring knowledge distillation for lightweight, low-latency self-verifying models.

TECHNICAL SKILLS

- **Programming Languages:** Python, JavaScript, C
- **Machine Learning & AI:** PyTorch, Hugging Face Transformers, scikit-learn, Deep Learning, Natural Language Processing (NLP)
- **Research Methods:** Model Fine-tuning, Transfer Learning, Adversarial Machine Learning, Performance Benchmarking, Reproducible Research
- **Scientific Computing:** NumPy, pandas
- **Tools & Platforms:** Docker, Git, GitHub, Hugging Face Spaces, LaTeX
- **Software Development:** React.js, Node.js, Express.js, MongoDB, MySQL, SQLite

LANGUAGE PROFICIENCY

- **English:** Full professional proficiency - Medium of Instruction (MOI) certified, BRAC University, 2026
- **Bengali:** Native

AWARDS & HONORS

- **Dean's List, BRAC University**, Fall 2024
- **Vice Chancellor's List, BRAC University**, Summer 2025
- **2nd Runner-up - CSE471 Project Demonstration**, BRAC University, 2025
- **Academic Distinction**, BSc in Computer Science and Engineering, BRAC University, 2026

REFERENCES

Dr. Muhammad Iqbal Hossain

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